

Day 18 — Tue 22 Sep

Countdown: 4 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🕒 GK — MCQ practice bank (self-test)

Source: *General/04_GK_MCQ_BANK.md*

GK MCQ Bank — Practice Questions for Paper 1 (General English & GK)

Exam: NPSC CTSE 2026, General English & GK (Combined) — 100 marks, 3 hours, **26 September 2026**. GK is 50 of the 100 marks. This file is a **self-test bank**: 160 original multiple-choice questions across every GK topic in the official syllabus, plus a current-affairs "what to look up" block. Pair it with [03_GK_CAPSULE.md](#) for the underlying facts and explanations — this file is for drilling recall speed.

How this paper actually tests GK — and how to use this file

- The syllabus line is: *"General knowledge comprising of current events, general science, mental ability, Indian history, Indian polity, Indian economy, Geography with combination of objective type and short descriptive answers — 50 Marks."* In practice (see the real 2025 paper) that means a long run of **single-best-answer, four-option MCQs** — some straightforward recall, some "which of these statements is true" / "which of i, ii, iii is correct" combination items — answered **in a grid on the answer booklet**, not written out.
- ⚠️ **No negative marking applies to the General paper — verify this on your actual 2026 question paper's instruction page before you rely on it.** The real 2025 General paper's cover page states *"There will be no negative marking"* in black and white, and this is a **different rule from the Technical MCQ paper**, which does negative-mark. Because the rule can only be confirmed from the paper you are actually sitting, treat "no negative marking" as *likely but not certified* until you read your own booklet's instructions on exam day — and if it does turn out to carry negative marking, do not blind-guess; only answer once you can eliminate at least one option.
- **Self-test discipline:** cover the answer key, attempt each section's block cold, then open the [<details>](#) key. Re-attempt only what you got wrong, a few days later.
- **Your structural edge:** you are a Naga sitting a Nagaland exam. The Nagaland & North-East section below is deliberately the largest block — a national candidate cannot match you here, so over-learn it.
- **Accuracy discipline in this file:** every fact that can change over time (current office-holders, district counts, scheme outlays, population figures) is marked ⚠️ **verify** rather than stated as fixed truth — re-check any such figure from a current source before writing it in the real exam. Nothing in this file is invented; where an exact current fact was needed and could not be verified, a question **stem** is given instead so you know what to look up.

SECTION 1 — INDIAN POLITY (25 questions, incl. Article 371A)

1. The Constitution of India was **adopted** by the Constituent Assembly on which date?
A) 15 August 1947 B) 26 January 1950 C) 26 November 1949 D) 9 December 1946
2. Which of the following words were added to the Preamble by the 42nd Amendment (1976)?
A) Sovereign, Democratic, Republic B) Socialist, Secular, Integrity
C) Justice, Liberty, Equality D) Federal, Secular, Unity
3. Dr. B. R. Ambedkar's famous description of Article 32 as "the heart and soul of the Constitution" refers to which right?
A) Right to Equality B) Right to Freedom of Religion
C) Right to Constitutional Remedies D) Right against Exploitation
4. Which Fundamental Right was removed by the 44th Amendment (1978) and made only a legal right under Article 300A?
A) Right to Education B) Right to Property C) Right to Vote D) Right to Privacy
5. Which writ is issued by a court to question a person's authority to hold a public office they are not entitled to?
A) Habeas Corpus B) Mandamus C) Quo Warranto D) Certiorari
6. The Directive Principles of State Policy (Part IV) are, unlike Fundamental Rights,
A) enforceable only in a High Court B) not enforceable by any court
C) enforceable only during an Emergency D) enforceable only against private citizens
7. Fundamental Duties were added to the Constitution on the recommendation of which committee?
A) Sarkaria Commission B) Swaran Singh Committee
C) Balwant Rai Mehta Committee D) Punchhi Commission
8. A Money Bill, under the Constitution, can be introduced only in which House?
A) Rajya Sabha B) Either House C) Lok Sabha D) A joint sitting of both Houses
9. Which body is described as the "guardian of the public purse" for auditing government accounts?
A) NITI Aayog B) Comptroller and Auditor General (CAG)
C) Finance Commission D) Reserve Bank of India
10. The "Basic Structure Doctrine" — that Parliament can amend the Constitution but not alter its basic structure — arose from which landmark case?
A) Golaknath case B) Minerva Mills case C) Kesavananda Bharati case D) Shah Bano case
11. A National Emergency under Article 352 can be proclaimed on the grounds of war, external aggression, or:
A) internal disturbance B) armed rebellion C) economic crisis D) a state's request
12. Which Fundamental Rights cannot be suspended even during a National Emergency, after the 44th Amendment?
A) Articles 14 and 19 B) Articles 20 and 21 C) Articles 25 and 26 D) Articles 32 and 226
13. NITI Aayog, which replaced the Planning Commission in 2015, is:
A) a constitutional body B) a statutory body created by an Act of Parliament
C) neither constitutional nor statutory — an executive policy body D) a part of the judiciary
14. Under the 73rd Amendment (1992), which is the lowest tier of rural local government?
A) Zilla Parishad B) Panchayat Samiti C) Gram Panchayat D) Municipal Council
15. Article 371A, which grants special constitutional provisions, applies to which state?
A) Mizoram B) Nagaland C) Manipur D) Sikkim
16. Article 371A was inserted into the Constitution by which amendment?
A) 7th Amendment B) 13th Amendment C) 42nd Amendment D) 61st Amendment

17. Article 371A took effect on which date — the day Nagaland became a state?
A) 21 January 1972 B) 1 December 1963 C) 26 January 1950 D) 20 February 1987
18. Under Article 371A(1)(a), no Act of Parliament applies to Nagaland in respect of Naga religious/social practice, customary law, land ownership, and one more matter, unless the state Assembly so resolves. What is that fourth protected matter?
A) Taxation of Naga businesses B) Administration of civil and criminal justice involving decisions according to Naga customary law
C) Recruitment to the state civil service D) The medium of instruction in schools
19. Article 371A gives the Governor of Nagaland a special personal responsibility (exercised in his individual judgement) with respect to:
A) education policy B) law and order, so long as internal disturbances continue
C) issuing currency D) appointing the Chief Minister
20. Which of the following North-Eastern states does NOT have Part IX (Panchayati Raj, 73rd Amendment) applying to it, under Article 243M?
A) Assam B) Tripura C) Nagaland D) Sikkim
21. Which Article deals with special provisions for Mizoram (not to be confused with Nagaland's 371A)?
A) 371B B) 371C C) 371F D) 371G
22. In Nagaland, in place of the standard Panchayati Raj structure, local governance is carried out chiefly through:
A) District Collectors appointed by the Centre B) traditional Village Councils and Village Development Boards (VDBs) C) directly elected mayors D) the Zilla Parishad system
23. The Finance Commission, which recommends the sharing of tax revenue between the Centre and the States, is constituted under which Article?
A) Article 280 B) Article 315 C) Article 324 D) Article 356
24. Which of the following is an accurate statement about the Rajya Sabha?
A) It can be dissolved by the President B) One-third of its members retire every two years, as it is a permanent House C) Money Bills can originate there
D) Its presiding officer is called the Speaker
25. The 101st Constitutional Amendment (2016) is associated with the introduction of:
A) the Right to Education B) the anti-defection law C) the Goods and Services Tax (GST)
D) the voting age reduction to 18

Answer key — Section 1 (Polity)

1. **C** — Adopted 26 Nov 1949; it *came into force* on 26 Jan 1950 (a classic trap: the two dates are often swapped in questions). 2. **B** — "Socialist, Secular, Integrity" were added; the original Preamble already had Sovereign/Democratic/Republic and Justice/Liberty/Equality/Fraternity. 3. **C** — Article 32 (Right to Constitutional Remedies) lets a citizen move the Supreme Court directly to enforce Fundamental Rights. 4. **B** — Right to Property was removed from Part III and demoted to a legal right under Art. 300A. 5. **C** — Quo Warranto = "by what authority" — challenges an unauthorised holding of public office. 6. **B** — DPSPs (Art. 37) are explicitly non-justiciable, unlike Fundamental Rights. 7. **B** — The Swaran Singh Committee recommended Fundamental Duties, added by the 42nd Amendment, inspired by the USSR constitution. 8. **C** — Money Bills (Art. 110) can be introduced only in the Lok Sabha; the Rajya Sabha can only recommend changes within 14 days. 9. **B** — The CAG (Art. 148) audits Union and State government accounts. 10. **C** — *Kesavananda Bharati v. State of Kerala* (1973) established the Basic Structure Doctrine. 11. **B** — The ground was changed from "internal disturbance" to "armed rebellion" by the 44th Amendment, to prevent misuse of Emergency powers (as happened in 1975). 12. **B** — Articles 20 (protection re conviction) and 21 (life & personal liberty) remain in force even in a National Emergency. 13. **C** — NITI Aayog is neither a constitutional body (not in the Constitution) nor a statutory body (no Act created it) — it is an executive resolution body, like the Planning Commission it replaced. 14. **C** — Gram Panchayat (village) → Panchayat Samiti (block) → Zilla Parishad (district) is the three-tier order; Gram Panchayat is the lowest. 15. **B** — Article 371A is Nagaland's own special-provisions article. 16. **B** —

Inserted by the 13th Constitutional Amendment Act, 1962. 17. **B** — 1 December 1963, the day Nagaland became India's 16th state. 18. **B** — The four protected subjects are: religious/social practice, Naga customary law, administration of justice per Naga customary law, and ownership/transfer of land and its resources. 19. **B** — The Governor's individual-judgement special responsibility under 371A concerns law and order during internal disturbances. 20. **C** — Article 243M excludes Nagaland (along with Meghalaya and Mizoram) from Part IX; Assam, Tripura and Sikkim are not among the states excluded. 21. **D** — Article 371G is Mizoram's special-provisions article (do not confuse with Nagaland's 371A). 22. **B** — Nagaland runs local governance through traditional Village Councils and VDBs, plus the Communitisation Act, 2002, rather than the standard Panchayati Raj structure. 23. **A** — The Finance Commission is constituted under Article 280, every five years. 24. **B** — The Rajya Sabha is a permanent body; one-third retire every two years. It cannot be dissolved, cannot originate Money Bills, and its presiding officer is the Vice-President (Chairman), not the Speaker. 25. **C** — The 101st Amendment (2016) introduced GST, effective 1 July 2017.

SECTION 2 — NAGALAND & NORTH-EAST GK (35 questions — your highest-yield block)

2A. Statehood, Article 371A context & administration (Q26–34)

26. Which early Naga political organisation, formed in 1918 at Kohima, is considered the starting point of modern Naga political consciousness?
A) Naga National Council B) Naga Club C) Naga People's Convention D) Naga Hills District Council
27. In 1929, the Naga Club submitted a memorandum to which British commission, asking that the Naga Hills be left out of the reforms proposed for the rest of India?
A) Cripps Mission B) Simon Commission C) Cabinet Mission D) Radcliffe Commission
28. The Naga National Council (NNC), a leading force in the Naga political movement, was formed in which year?
A) 1929 B) 1946 C) 1957 D) 1963
29. The **16-Point Agreement**, the political basis on which Nagaland statehood was negotiated, was signed in which year, between the Naga People's Convention and the Government of India?
A) 1957 B) 1960 C) 1962 D) 1963
30. Nagaland became India's 16th state on 1 December 1963. Which act of Parliament formally created the state?
A) State of Nagaland Act, 1962 B) Government of India Act, 1935
C) North-Eastern Areas (Reorganisation) Act, 1971 D) Assam Reorganisation Act, 1963
31. The Nagaland Legislative Assembly has how many seats?
A) 40 B) 60 C) 90 D) 126
32. How many seats does Nagaland send to the Lok Sabha?
A) One B) Two C) Three D) Five
33. Which of the following is the official language of the state of Nagaland?
A) Nagamese B) Hindi C) English D) Ao
34. What is "Nagamese", widely spoken as a link language across Nagaland's many tribes?
A) A dialect of Hindi spoken only in Dimapur B) An Assamese-based creole/pidgin used as a common lingua franca C) The official script used for all Naga languages
D) A Tibeto-Burman classical language taught in schools

2B. State profile & geography (Q35–46)

35. What is the capital of Nagaland?
A) Dimapur B) Kohima C) Mokokchung D) Wokha
36. Which of the following is Nagaland's largest commercial town and its principal transport hub (rail and air)? [△ verify](#) — infrastructure status can change
A) Kohima B) Dimapur C) Tuensang D) Zunheboto
37. Which country shares an international border with Nagaland on its eastern side?
A) Bangladesh B) Bhutan C) Myanmar D) China
38. Which of the following states does NOT share a border with Nagaland?
A) Assam B) Manipur C) Arunachal Pradesh D) Meghalaya
39. What is the highest peak of Nagaland, located on the Myanmar border in Kiphire district?
A) Japfü peak B) Mount Saramati C) Dzükou peak D) Mount Tempu
40. The Dzükou Valley, famous for its seasonal lily and trekking trail, lies on the border between Nagaland and which neighbouring state?
A) Assam B) Manipur C) Arunachal Pradesh D) Meghalaya
41. Which is Nagaland's only national park, located in Peren district? [△ verify](#) — protected-area status/count can be updated
A) Fakim Wildlife Sanctuary B) Intanki National Park
C) Keibul Lamjao National Park D) Kaziranga National Park
42. Which traditional farming method — clear, burn, cultivate for a season or two, then leave fallow — is widely practised in the hills of Nagaland?
A) Terrace farming only B) Jhum (shifting) cultivation C) Plantation farming D) Contour ploughing
43. Which river system does Nagaland's Tizu river drain into, distinguishing it from most of the state's other rivers?
A) The Brahmaputra, like the Doyang and Dhansiri B) The Chindwin (in Myanmar), draining eastward
C) The Ganga D) It is an endorheic (closed) river with no outlet
44. What is the state animal of Nagaland, a semi-domesticated bovine seen as a traditional symbol of wealth and status?
A) Hoolock gibbon B) Mithun C) Clouded leopard D) Blyth's tragopan
45. What is the state bird of Nagaland?
A) Great hornbill B) Blyth's tragopan C) Peacock D) Kalij pheasant
46. The number of administrative districts in Nagaland has increased in recent years with the creation of new districts. [△ verify](#) the current exact count before quoting it in the exam — roughly how would you correctly hedge this in a written answer?
A) State a specific number confidently even if unsure B) State that the number "has grown into the high teens with several newly carved districts" and verify the exact figure from a current source
C) Refuse to answer D) Assume it equals the number of Lok Sabha seats

2C. Tribes, festivals & culture (Q47–55)

47. The **Sekrenyi** festival, celebrated in February, is the signature festival of which Naga tribe, associated with the Kohima region and terraced wet-rice cultivation?
A) Ao B) Angami C) Konyak D) Sumi
48. The **Moatsü** festival, celebrated around May, belongs to which tribe, historically among the first Naga groups to receive Western education and Christianity?
A) Lotha B) Ao C) Chakhesang D) Chang
49. **Tuluni**, a mid-year harvest festival, is associated with which tribe of the Zunheboto region?
A) Sumi (Sema) B) Rengma C) Pochury D) Phom
50. **Tokhü Emong**, a post-harvest festival celebrated around November, is the signature festival of which tribe of the Wokha region?

Kohima is the capital. 36. **B** — Dimapur is the largest commercial town and the state's main rail/air gateway. $\triangle$ verify current infrastructure details before quoting specifics. 37. **C** — Myanmar borders Nagaland to the east. 38. **D** — Nagaland's neighbours are Assam, Arunachal Pradesh, Manipur, and Myanmar; Meghalaya does not border it. 39. **B** — Mount Saramati (~3,841 m $\triangle$ verify) is Nagaland's highest peak, on the Myanmar border. 40. **B** — Dzükou Valley straddles the Nagaland–Manipur border, near Kohima. 41. **B** — Intanki National Park, in Peren district, is commonly cited as Nagaland's only national park. $\triangle$ verify sole status and boundaries before an exam answer. 42. **B** — Jhum (shifting/slash-and-burn) cultivation is the traditional hill farming method. 43. **B** — Unlike the Doyang and Dhansiri (which drain west to the Brahmaputra), the Tizu river drains eastward into the Chindwin in Myanmar — a classic discriminator fact. 44. **B** — Mithun (*Bos frontalis*) is the state animal. 45. **B** — Blyth's Tragopan is the state bird. 46. **B** — The correct exam technique for any volatile count (district numbers, etc.) is qualitative hedging plus a verify-before-writing habit, never a confidently stated unverified figure. 47. **B** — Sekrenyi (Phousanyi), February, is the Angami tribe's signature festival. 48. **B** — Moatsü, May, belongs to the Ao tribe (Mokokchung region), historically early adopters of Western education and Christianity. 49. **A** — Tuluni, July, is a Sumi (Sema) festival of the Zunheboto region. 50. **B** — Tokhü Emong, November, is a Lotha festival of the Wokha region. 51. **A** — Aoleang Monyu, April, is the Konyak tribe's major festival; Konyaks are historically associated with Mon district and hereditary *Anghs*. 52. **B** — Chakhesang, a composite identity from the Phek region. 53. **B** — Naga Heritage Village, Kisama, near Kohima — not Kohima town itself. 54. **C** — The Hornbill Festival began in 2000. 55. **B** — The hornbill is revered in Naga folklore and features in traditional attire/dance, hence lending its name to the festival. 56. **B** — The Battle of Kohima (1944) was a decisive WWII engagement that helped halt the Japanese advance into India. 57. **B** — The Kohima War Cemetery commemorates Allied soldiers who died in the 1944 battle. 58. **B** — The Naga King Chilli has a GI tag. $\triangle$ verify current GI status/details before writing a specific claim. 59. **B** — Imphal is Manipur's capital. 60. **C** — Sikkim is commonly described as the "one brother" joining the Seven Sisters (Assam, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, Arunachal Pradesh) to make up the eight North-Eastern states.

SECTION 3 — INDIAN HISTORY (20 questions)

61. Which ancient civilisation is associated with the planned cities of Harappa, Mohenjo-daro, Dholavira and Lothal, and the still-undeciphered script?
A) Vedic civilisation B) Indus Valley (Harappan) civilisation
C) Mauryan civilisation D) Gupta civilisation
62. Which of the four Vedas is generally regarded as the oldest?
A) Sama Veda B) Yajur Veda C) Rig Veda D) Atharva Veda
63. Chandragupta Maurya founded the Mauryan Empire with the guidance of which strategist, the author of the *Arthashastra*?
A) Chanakya (Kautilya) B) Ashoka C) Bindusara D) Megasthenes
64. Emperor Ashoka's conversion to Buddhism is traditionally linked to his remorse after which military campaign?
A) The Battle of Panipat B) The Kalinga War C) The conquest of Magadha D) The Sindh campaign
65. The Gupta period is often called the "Golden Age" of ancient India, associated with scholars such as Aryabhata and Kalidasa, under which ruler particularly?
A) Chandragupta Maurya B) Harshavardhana C) Chandragupta II (Vikramaditya) D) Samudragupta
66. Babur established Mughal rule in India after defeating Ibrahim Lodi at which battle in 1526?
A) Battle of Plassey B) Battle of Panipat (First) C) Battle of Haldighati D) Battle of Buxar
67. Which Mughal emperor commissioned the construction of the Taj Mahal?
A) Akbar B) Jahangir C) Shah Jahan D) Aurangzeb
68. Which historical figure founded the Maratha Empire and is remembered for his use of guerrilla warfare tactics against the Mughals?

- A) Shivaji B) Bajirao I C) Sambhaji D) Balaji Vishwanath
69. The Battle of Plassey (1757), which marked the beginning of British political power in Bengal, was fought primarily between the British East India Company and:
A) Tipu Sultan B) Siraj-ud-Daulah, the Nawab of Bengal C) the Marathas D) the Mughal emperor directly
70. The Revolt of 1857 is also widely known by which other name, referring to the role of Indian soldiers in the British army?
A) The Sepoy Mutiny B) The Quit India Movement C) The Salt Satyagraha D) The Non-Cooperation Movement
71. The Indian National Congress was founded in 1885 with the involvement of which retired British civil servant?
A) Lord Curzon B) A. O. Hume C) Lord Ripon D) Lord Dalhousie
72. Lord Curzon's 1905 Partition of Bengal directly triggered which movement, urging the boycott of British goods and promotion of indigenous industry?
A) Non-Cooperation Movement B) Swadeshi Movement C) Quit India Movement D) Civil Disobedience Movement
73. The Jallianwala Bagh massacre of 13 April 1919 occurred amid protests against which repressive legislation?
A) The Rowlatt Act B) The Government of India Act, 1935 C) The Charter Act D) The Indian Councils Act
74. Mahatma Gandhi's Non-Cooperation Movement (1920–22) was called off after which violent incident?
A) The Jallianwala Bagh massacre B) The Chauri Chaura incident C) The Simon Commission protests D) The Quit India arrests
75. The 1930 Dandi Salt March, which launched the Civil Disobedience Movement, was led by Gandhi from Sabarmati Ashram to which coastal village?
A) Dandi B) Porbandar C) Surat D) Daman
76. The "Quit India Movement" of 1942 was launched with which famous call to action by Gandhi?
A) "Swaraj is my birthright" B) "Do or Die" C) "Inquilab Zindabad" D) "Satyameva Jayate"
77. Subhas Chandra Bose is chiefly remembered for founding/leading which armed organisation seeking India's independence during the Second World War?
A) The Ghadar Party B) The Indian National Army (Azad Hind Fauj) C) The Home Rule League D) The Hindustan Socialist Republican Association
78. India became independent, with the transfer of power and Partition, on which date?
A) 26 January 1950 B) 15 August 1947 C) 2 October 1947 D) 30 January 1948
79. Who is credited with the phrase "Swaraj is my birthright and I shall have it"?
A) Mahatma Gandhi B) Bal Gangadhar Tilak C) Jawaharlal Nehru D) Subhas Chandra Bose
80. Sardar Vallabhbhai Patel, called the "Iron Man of India," is chiefly remembered for his role in:
A) drafting the Constitution B) the integration of the princely states into the Indian Union C) leading the Quit India Movement D) founding the Indian National Army

Answer key — Section 3 (History)

61. ****B**** — Indus Valley/Harappan civilisation, Bronze Age, planned cities, undeciphered script. 62. ****C**** — The Rig Veda is generally considered the oldest of the four Vedas. 63. ****A**** — Chanakya (Kautilya), author of the **Arthashastra**, guided Chandragupta Maurya. 64. ****B**** — The bloodshed of the Kalinga War is traditionally linked to Ashoka's turn to Buddhism. 65. ****C**** — Chandragupta II (Vikramaditya) presided over the height of Gupta cultural flourishing; Aryabhata and Kalidasa are associated with this era broadly. 66. ****B**** — The First Battle of Panipat (1526), where Babur defeated Ibrahim Lodi, established Mughal rule. 67. ****C**** — Shah Jahan commissioned the Taj Mahal. 68. ****A**** — Shivaji founded the Maratha Empire and is known for guerrilla warfare. 69. ****B**** — Siraj-ud-Daulah, Nawab of Bengal, was defeated by the British at Plassey

(1757), aided by the betrayal of Mir Jafar. 70. **A** — Also called the "Sepoy Mutiny," referencing the Indian soldiers (sepoys) who revolted; historians also call it the "First War of Indian Independence." 71. **B** — A. O. Hume, a retired British civil servant, was instrumental in founding the INC in 1885. 72. **B** — The 1905 Partition of Bengal triggered the Swadeshi Movement. 73. **A** — The Rowlatt Act's repressive provisions were being protested when the Jallianwala Bagh massacre occurred. 74. **B** — The Chauri Chaura incident (1922), where protesters killed policemen, led Gandhi to call off Non-Cooperation. 75. **A** — Dandi, the coastal village where Gandhi broke the salt law, ending the 24-day march from Sabarmati Ashram. 76. **B** — "Do or Die" was Gandhi's call at the launch of the Quit India Movement in 1942. 77. **B** — The Indian National Army (INA/Azad Hind Fauj), which Bose led, fought alongside Japan against British forces. 78. **B** — 15 August 1947. 79. **B** — Bal Gangadhar Tilak. 80. **B** — Sardar Patel is remembered chiefly for integrating over 500 princely states into the Union after independence.

SECTION 4 — INDIAN ECONOMY (20 questions)

81. GDP (Gross Domestic Product) measures the value of goods and services:
 - A) produced by a country's citizens anywhere in the world
 - B) produced within a country's geographical borders in a given year
 - C) consumed by government agencies only
 - D) exported by a country in a year
82. GNP (Gross National Product) differs from GDP by additionally accounting for:
 - A) government subsidies
 - B) net income earned by nationals from abroad
 - C) foreign direct investment inflows only
 - D) inflation adjustment
83. A sustained rise in the general price level of an economy is called:
 - A) deflation
 - B) inflation
 - C) stagflation only
 - D) recession
84. "Cost-push" inflation is primarily driven by:
 - A) excess money chasing too few goods
 - B) rising costs of inputs (e.g. raw materials, wages) pushing up final prices
 - C) a fall in the money supply
 - D) currency appreciation
85. In India, retail-level inflation is most commonly measured using which index?
 - A) Wholesale Price Index (WPI)
 - B) Consumer Price Index (CPI)
 - C) Human Development Index (HDI)
 - D) Index of Industrial Production (IIP)
86. Fiscal policy — taxation and government spending — is primarily the domain of:
 - A) the Reserve Bank of India
 - B) the Government (Ministry of Finance)
 - C) the Securities and Exchange Board of India
 - D) the World Bank
87. Monetary policy — controlling money supply and interest rates — is primarily the domain of:
 - A) the Government's Ministry of Finance
 - B) the Reserve Bank of India (RBI)
 - C) the Election Commission
 - D) NITI Aayog
88. Raising the **repo rate** (the rate at which RBI lends to commercial banks) is a tool typically used to:
 - A) encourage more borrowing and boost demand
 - B) make borrowing costlier and help cool inflation
 - C) directly control the fiscal deficit
 - D) fix the exchange rate
89. The Union Budget is presented under which Article of the Constitution, as the "Annual Financial Statement"?
 - A) Article 110
 - B) Article 112
 - C) Article 280
 - D) Article 360
90. The Reserve Bank of India, India's central bank, was established in which year?
 - A) 1926
 - B) 1935
 - C) 1947
 - D) 1969
91. GST (Goods and Services Tax), described as "One Nation, One Tax," came into effect in India on which date?
 - A) 1 April 2016
 - B) 1 July 2017
 - C) 26 January 2017
 - D) 1 January 2018
92. Under the GST framework, tax collected on an intra-state sale is split into which two components?

- A) IGST and Customs Duty B) CGST and SGST C) VAT and Excise D) Cess and Surcharge
93. Which body, chaired by the Union Finance Minister, decides GST rates and rules?
A) SEBI B) The GST Council C) NITI Aayog D) The Finance Commission
94. NITI Aayog, established in 2015, replaced which earlier planning body?
A) The Finance Commission B) The Planning Commission C) The National Development Council D) The Tariff Commission
95. MGNREGA is a rural employment guarantee scheme that guarantees how many days of wage employment per year to a rural household? Δ verify current scheme parameters
A) 50 days B) 100 days C) 150 days D) 200 days
96. Which government scheme is primarily focused on financial inclusion — opening basic bank accounts for the unbanked population?
A) Ayushman Bharat B) PM Jan Dhan Yojana C) Skill India D) Digital India
97. India's "Act East Policy," which promotes engagement with Southeast Asian nations, holds particular relevance for which part of India, given its geography and Myanmar border?
A) The western coastal states B) The North-Eastern states, including Nagaland C) The Himalayan states only D) The southern peninsula
98. In national income accounting, "Net National Product" (NNP) is derived by subtracting which item from GNP?
A) Taxes B) Depreciation C) Subsidies D) Exports
99. The Finance Commission's core function is to recommend the:
A) annual Union Budget allocations B) sharing/devolution of tax revenues between the Centre and the States C) repo rate for the year D) list of GST-exempt goods
100. Per capita income is calculated as:
A) Total national income $\times$ population B) Total national income $\div$ population C) GDP – depreciation D) GDP + net foreign income

Answer key — Section 4 (Economy)

81. **B** — GDP is production within national borders, regardless of who produces it. 82. **B** — GNP = GDP + net income earned by the country's nationals from abroad. 83. **B** — Inflation is a sustained rise in the general price level. 84. **B** — Cost-push inflation comes from rising input costs; demand-pull comes from excess demand. 85. **B** — CPI is India's primary retail inflation gauge (used for inflation targeting); WPI tracks wholesale prices. 86. **B** — Fiscal policy (taxation, spending, borrowing) is run by the Government/Ministry of Finance. 87. **B** — Monetary policy (money supply, interest rates) is run by the RBI. 88. **B** — A higher repo rate raises borrowing costs economy-wide, cooling demand and inflation. 89. **B** — Article 112 is the constitutional basis for the Union Budget ("Annual Financial Statement"). 90. **B** — RBI was established in 1935. 91. **B** — GST came into force on 1 July 2017 (enabled by the 101st Amendment, 2016). 92. **B** — An intra-state sale splits tax into CGST (Centre's share) and SGST (State's share); IGST applies to inter-state sales. 93. **B** — The GST Council, chaired by the Union Finance Minister with state representation, decides GST matters. 94. **B** — NITI Aayog replaced the Planning Commission in 2015. 95. **B** — MGNREGA's headline guarantee is 100 days of wage employment per rural household per year. Δ verify current parameters/any revisions before quoting in the exam. 96. **B** — PM Jan Dhan Yojana is the flagship financial-inclusion scheme (basic bank accounts). 97. **B** — The North-Eastern states, bordering Myanmar, are the geographic gateway for the Act East Policy — directly relevant to Nagaland. 98. **B** — NNP = GNP – depreciation. 99. **B** — The Finance Commission (Art. 280) recommends Centre-State tax devolution every five years. 100. **B** — Per capita income = total national income $\div$ population.

SECTION 5 — GEOGRAPHY (20 questions)

101. Which of the following is the highest mountain range in India, forming its

northern boundary?

A) The Aravallis B) The Vindhyas C) The Himalayas D) The Western Ghats

102. India's highest peak (entirely within Indian territory) is:

A) Mount Everest B) Kanchenjunga C) Nanda Devi D) K2

103. Which of the following rivers is snow-fed and perennial, unlike the seasonal peninsular rivers?

A) Godavari B) Krishna C) Ganga D) Kaveri

104. Which is the longest river confined entirely to the Indian peninsula (south of the Narmada), sometimes called the "Dakshin Ganga"?

A) Krishna B) Kaveri C) Godavari D) Tapi

105. Which of the following are the two major west-flowing rivers of peninsular India that drain into the Arabian Sea, unlike most other peninsular rivers?

A) Godavari and Krishna B) Kaveri and Krishna C) Narmada and Tapi D) Mahanadi and Brahmani

106. Which monsoon system delivers the bulk of India's annual rainfall, typically between June and September?

A) The North-East (retreating) Monsoon B) The South-West Monsoon
C) The Western Disturbance D) The Loo winds

107. Which type of soil, found in the river plains, is the most widespread and agriculturally significant soil type in India?

A) Black/Regur soil B) Red soil C) Alluvial soil D) Laterite soil

108. Black (Regur) soil, ideal for growing cotton, is chiefly found in which region?

A) The Indo-Gangetic plains B) The Deccan plateau C) The Thar Desert D) The Himalayan foothills

109. Which of the following is Asia's oldest oil refinery, located in Assam?

A) Barauni refinery B) Digboi refinery C) Jamnagar refinery D) Mathura refinery

110. Which desert, located primarily in Rajasthan, is India's major arid region?

A) The Thar Desert B) The Kutch Desert C) The Rann Desert D) The Ladakh Desert

111. The Western and Eastern Ghats are mountain ranges bordering which physical feature of peninsular India?

A) The Northern Plains B) The Deccan Plateau (its western and eastern edges)
C) The Thar Desert D) The Himalayas

112. Which of the following states shares an international border with China?

A) Nagaland B) Arunachal Pradesh C) Manipur D) Tripura

113. What is the capital of Assam, a state bordering Nagaland to the west?

A) Guwahati B) Dispur C) Silchar D) Tezpur

114. Kaziranga National Park, famous for the one-horned rhinoceros, is located in which state?

A) Assam B) Nagaland C) Meghalaya D) Arunachal Pradesh

115. Which two towns in Meghalaya are frequently cited among the wettest places on Earth?

A) Shillong and Tura B) Cherrapunji and Mawsynram C) Jowai and Nongstoin D) Baghmara and Williamnagar

116. Sikkim, which borders Nagaland's neighbourhood only distantly but is notable in NE geography, is recognised as India's:

A) largest state by area B) first fully organic state
C) only landlocked Union Territory D) only state without a Legislative Assembly

117. What is a distinguishing geographic feature of Loktak Lake in Manipur, home to the Keibul Lamjao National Park?

A) It is India's largest saltwater lake B) It contains floating masses of vegetation called *phumdis*, supporting the only floating national park
C) It is entirely man-made D) It lies above 5,000 metres altitude

118. In terms of physical divisions, the Northern Plains of India are primarily formed by which process?
 A) Volcanic activity B) Alluvial deposition by rivers (Indo-Gangetic-Brahmaputra system)
 C) Glacial erosion D) Coral reef formation
119. Which of the following best describes "jhum" cultivation, common in Nagaland and other NE hill states?
 A) Large-scale mechanised single-crop farming B) Shifting/slash-and-burn cultivation on hill slopes, followed by a fallow period C) Floating-bed cultivation on a lake
 D) Terrace irrigation exclusively for rice
120. Laterite soil, found in areas of heavy rainfall (including parts of the North-East and Western Ghats), is generally:
 A) rich in humus and ideal for wheat B) leached of nutrients and less fertile, though suited to certain plantation crops C) found only in deserts D) the same as alluvial soil

Answer key — Section 5 (Geography)

101. ****C**** — The Himalayas form India's northern mountain boundary. 102. ****B**** — Kanchenjunga (Sikkim) is the highest peak wholly within India; Everest is higher overall but not in Indian territory. 103. ****C**** — The Ganga (like the Yamuna, Brahmaputra, Indus) is Himalayan-fed and perennial; peninsular rivers are rain-fed and seasonal. 104. ****C**** — The Godavari, called the "Dakshin Ganga," is the longest peninsular river. 105. ****C**** — Narmada and Tapi flow west into the Arabian Sea; most other peninsular rivers flow east into the Bay of Bengal. 106. ****B**** — The South-West Monsoon (June–September) brings the majority of India's rainfall. 107. ****C**** — Alluvial soil, in the river plains, is the most widespread and fertile soil type. 108. ****B**** — Black/Regur soil is characteristic of the Deccan plateau and is well-suited to cotton. 109. ****B**** — Digboi refinery, Assam, is Asia's oldest oil refinery. 110. ****A**** — The Thar Desert, mainly in Rajasthan, is India's principal arid zone. 111. ****B**** — The Western and Eastern Ghats flank the Deccan Plateau on its west and east. 112. ****B**** — Arunachal Pradesh shares an international border with China (Nagaland's neighbours are Assam, Arunachal Pradesh, Manipur and Myanmar — not China). 113. ****B**** — Dispur is Assam's capital; Guwahati is the largest city but not the capital — a classic trap. 114. ****A**** — Kaziranga National Park is in Assam. 115. ****B**** — Cherrapunji (Sohra) and Mawsynram, both in Meghalaya, are frequently cited as among the wettest places on Earth. 116. ****B**** — Sikkim is India's first fully organic state. 117. ****B**** — Loktak Lake's floating *phumdis* (vegetation mats) support Keibul Lamjao, the world's only floating national park. 118. ****B**** — The Northern Plains are alluvial plains built by the Ganga-Yamuna-Brahmaputra river system. 119. ****B**** — Jhum is shifting/slash-and-burn hill cultivation with a fallow recovery period, common in Nagaland and the wider North-East. 120. ****B**** — Laterite soils form under heavy rainfall/leaching, are relatively nutrient-poor for staple grains, but support certain plantation crops (tea, cashew, rubber).

SECTION 6 — GENERAL SCIENCE (20 questions)

121. What is the SI unit of force?
 A) Joule B) Newton C) Watt D) Pascal
122. Newton's First Law of Motion is also known as the law of:
 A) Action-reaction B) Inertia C) Conservation of momentum D) Gravitation
123. According to Newton's Second Law, force equals:
 A) mass ÷ acceleration B) mass × acceleration C) mass × velocity D) mass ÷ velocity
124. Sound is a wave that requires a medium to travel and therefore cannot travel through:
 A) water B) air C) a vacuum D) a solid
125. The approximate audible frequency range for a healthy human ear is:
 A) 2 Hz – 200 Hz B) 20 Hz – 20,000 Hz C) 200 Hz – 2,000 Hz D) 20,000 Hz – 200,000 Hz
126. A convex lens, as commonly used in a magnifying glass, causes light rays to:
 A) diverge B) converge C) travel in a straight line unaffected D) reflect completely

127. Ohm's Law relating voltage, current and resistance is expressed as:
A) $P = VI$ B) $V = IR$ C) $F = ma$ D) $E = mc^2$
128. An acid, when tested with litmus paper, characteristically:
A) turns blue litmus red B) turns red litmus blue C) has no effect on litmus D) turns litmus green
129. Pure water is considered chemically neutral because its pH value is:
A) 0 B) 7 C) 14 D) negative
130. What is the chemical name commonly used for baking soda?
A) Sodium chloride B) Sodium bicarbonate C) Calcium oxide D) Sodium carbonate
131. The rusting of iron is a chemical process that essentially requires the presence of:
A) only oxygen B) oxygen and moisture (water) C) only carbon dioxide D) only heat, no oxygen
132. In atomic structure, the "atomic number" of an element refers to the number of:
A) neutrons in the nucleus B) protons in the nucleus
C) electrons in the outermost shell only D) protons plus neutrons
133. Isotopes of an element have the same number of protons but differ in their number of:
A) electrons B) neutrons C) protons D) valence shells
134. Which vitamin's deficiency classically causes scurvy?
A) Vitamin A B) Vitamin B12 C) Vitamin C D) Vitamin D
135. Deficiency of which vitamin in children classically causes rickets, a bone disorder?
A) Vitamin A B) Vitamin C C) Vitamin D D) Vitamin K
136. Malaria, a disease affecting many parts of India, is caused by a protozoan parasite transmitted by which insect vector?
A) Aedes mosquito B) Female Anopheles mosquito C) Housefly D) Tsetse fly
137. Tuberculosis (TB) is caused by which type of pathogen?
A) A virus B) A bacterium (*Mycobacterium tuberculosis*) C) A protozoan D) A fungus
138. Photosynthesis, the process by which green plants make their food, uses carbon dioxide, water and sunlight to produce glucose and release which gas?
A) Nitrogen B) Oxygen C) Carbon monoxide D) Hydrogen
139. Red blood cells (RBCs) carry oxygen through the bloodstream primarily using which protein?
A) Insulin B) Haemoglobin C) Keratin D) Collagen
140. Which of the following correctly orders the classification of living organisms into kingdoms, as commonly taught (five-kingdom classification)?
A) Monera, Protista, Fungi, Plantae, Animalia B) Bacteria, Virus, Fungi, Plantae, Animalia
C) Protista, Monera, Algae, Plantae, Animalia D) Fungi, Monera, Virus, Plantae, Animalia

Answer key — Section 6 (General Science)

121. ****B**** — Newton (N) is the SI unit of force. 122. ****B**** — The First Law (inertia) states a body remains at rest or in uniform motion unless acted upon by an external force. 123. ****B**** — $F = ma$ (Second Law). 124. ****C**** — Sound is a mechanical wave and cannot propagate through a vacuum, unlike light. 125. ****B**** — 20 Hz to 20,000 Hz is the typical human audible range. 126. ****B**** — A convex lens converges light rays (used in magnifying glasses and to correct long-sightedness). 127. ****B**** — Ohm's Law: $V = IR$. 128. ****A**** — Acids turn blue litmus red; bases turn red litmus blue. 129. ****B**** — Pure water has pH 7, the neutral point. 130. ****B**** — Baking soda is sodium bicarbonate (NaHCO_3); washing soda is sodium carbonate (Na_2CO_3) — a common mix-up. 131. ****B**** — Rusting (formation of hydrated iron oxide) requires both oxygen and moisture; neither alone is sufficient. 132. ****B**** — Atomic number = number of protons; mass number = protons + neutrons. 133. ****B**** — Isotopes share proton count (same element) but differ in neutron count. 134. ****C**** — Vitamin C deficiency causes scurvy. 135. ****C**** — Vitamin D deficiency in children causes rickets (osteomalacia in adults). 136. ****B**** — Malaria is spread by the bite of the female Anopheles mosquito,

carrying the Plasmodium parasite. 137. ****B**** — TB is caused by the bacterium *Mycobacterium tuberculosis*. 138. ****B**** — Photosynthesis releases oxygen as a byproduct while producing glucose. 139. ****B**** — Haemoglobin in RBCs binds and transports oxygen. 140. ****A**** — The standard five-kingdom classification is Monera, Protista, Fungi, Plantae, Animalia.

SECTION 7 — MENTAL ABILITY (20 questions)

141. Look at this series: 4, 8, 14, 22, 32, ? — What number should come next?
A) 40 B) 42 C) 44 D) 46
142. Look at this series: 2, 6, 18, 54, ? — What number comes next?
A) 108 B) 118 C) 152 D) 162
143. If in a certain code, CAT is written as DBU (each letter shifted forward by one), how would DOG be written using the same rule?
A) EPH B) EPI C) FPH D) EOI
144. If in a certain code SUN is written as TVO (each letter +1), how would MOON be written?
A) NPPO B) NPOO C) NPPP D) OPQO
145. Pointing to a photograph, Raj said, "She is the daughter of my grandfather's only son." Raj is that son, and he has no brothers. How is the woman in the photograph related to Raj?
A) His mother B) His sister C) His daughter D) His cousin
146. A is the brother of B. C is the mother of A. D is C's father. How is B related to D?
A) Son/daughter B) Grandson/granddaughter C) Nephew/niece D) Father/mother
147. Facing north, a man walks 6 km, then turns right and walks 8 km. What is his straight-line distance from the starting point?
A) 8 km B) 10 km C) 12 km D) 14 km
148. A man starts facing South, turns 90° clockwise, then another 90° clockwise. Which direction does he now face?
A) North B) East C) South D) West
149. All squares are rectangles. All rectangles are quadrilaterals. Which conclusion necessarily follows?
A) All quadrilaterals are squares B) All squares are quadrilaterals
C) No square is a quadrilateral D) Some quadrilaterals are not rectangles
150. Complete the analogy: Pen : Write :: Knife : ?
A) Sharp B) Cut C) Kitchen D) Steel
151. Complete the analogy: Puppy : Dog :: Calf : ?
A) Cow B) Sheep C) Horse D) Goat
152. A sum of ₹900 is divided between two people in the ratio 4:5. What is the smaller share?
A) ₹360 B) ₹400 C) ₹450 D) ₹500
153. What is 30% of 250?
A) 65 B) 70 C) 75 D) 80
154. The cost price of an article is ₹600 and it is sold for ₹750. What is the profit percentage?
A) 20% B) 25% C) 30% D) 15%
155. A can complete a piece of work in 8 days, and B can complete the same work in 12 days. Working together, in how many days will they complete the work?
A) 4.0 days B) 4.8 days C) 5.0 days D) 6.0 days
156. A car travels at a constant speed of 90 km/h. What is this speed in metres per

second?

A) 20 m/s B) 25 m/s C) 30 m/s D) 36 m/s

157. What is the average of the numbers 12, 18, 24, and 30?

A) 20 B) 21 C) 22 D) 24

158. What is the simple interest on ₹5,000 at 8% per annum for 3 years?

A) ₹1,000 B) ₹1,100 C) ₹1,200 D) ₹1,400

159. A bag contains 4 red balls and 6 green balls. If one ball is drawn at random, what is the probability that it is red?

A) $\frac{2}{5}$ B) $\frac{3}{5}$ C) $\frac{1}{2}$ D) $\frac{1}{3}$

160. What is the angle between the hour hand and minute hand of a clock at exactly 6:00?

A) 90° B) 120° C) 150° D) 180°

Answer key — Section 7 (Mental Ability)

141. **C** — Differences are 4, 6, 8, 10 → next difference is 12 → $32 + 12 = 44$. 142. **D** — Each term is $\times 3$: 2, 6, 18, 54, **162**. 143. **A** — Each letter shifted +1: D→E, O→P, G→H → **EPH**. 144. **A** — M→N, O→P, O→P, N→O → **NPPO**. 145. **B** — Raj's grandfather's only son is Raj's own father (since Raj has no brothers, "my grandfather's only son" identifies the father through Raj himself). The daughter of Raj's father is Raj's sister — **B**. 146. **B** — C is A's mother, D is C's father, so D is A's grandfather. B, being A's sibling, is also D's grandchild → **grandson/granddaughter**. 147. **B** — Right-angle triangle with legs 6 and 8 → hypotenuse = $\sqrt{(36+64)} = \sqrt{100} = \mathbf{10\ km}$. 148. **A** — South, turn 90° clockwise → West; another 90° clockwise → **North**. 149. **B** — By transitivity: All squares are rectangles, all rectangles are quadrilaterals → **all squares are quadrilaterals**. 150. **B** — A pen's function is to write; a knife's function is to **cut** (tool : primary function). 151. **A** — A puppy grows into a dog; a calf grows into a **cow** (young : adult, cattle). 152. **B** — Total parts = 9, one part = 100 → smaller share (4 parts) = **₹400**. 153. **C** — 30% of 250 = $0.30 \times 250 = \mathbf{75}$. 154. **B** — Profit = $750 - 600 = 150$; profit% = $150/600 \times 100 = \mathbf{25\%}$. 155. **B** — Combined rate = $\frac{1}{8} + \frac{1}{12} = \frac{3}{24} + \frac{2}{24} = \frac{5}{24}$ per day → time = $24/5 = \mathbf{4.8\ days}$. 156. **B** — $90 \times \frac{5}{18} = \mathbf{25\ m/s}$. 157. **B** — $(12+18+24+30)/4 = 84/4 = \mathbf{21}$. 158. **C** — SI = $(5000 \times 8 \times 3)/100 = \mathbf{₹1,200}$. 159. **A** — $P(\text{red}) = \frac{4}{(4+6)} = \frac{4}{10} = \mathbf{\frac{2}{5}}$. 160. **D** — At 6:00, the hour hand is at 6 and the minute hand at 12, exactly opposite → **180°** .

SECTION 8 — CURRENT AFFAIRS: WHAT TO LOOK UP, NOT WHAT TO MEMORISE FROM HERE

This file will not invent current-affairs facts, names, or dates. The real 2025 General paper devoted a large opening block to events from the preceding 12 months (diplomatic visits, treaties, science/tech news, sports, awards) — so expect the 2026 paper to do the same for the window **September 2025 to September 2026**. Below are **question stems only** — the *shape* of what will likely be asked — each pointing to a topic you must verify from a live, reliable source (PIB, The Hindu, Nagaland Post/ Morung Express/Eastern Mirror for the NE angle) close to the exam date. Do **not** fill in specific names, dates or numbers from memory unless you have just verified them.

- [] **verify**: Who currently holds the offices of Governor and Chief Minister of Nagaland, and has either changed in the last 12 months?
- [] **verify**: What was the theme, dates, and any partner state/country of the most recent Hornbill Festival?
- [] **verify**: Has Nagaland gained any new district(s), and what is the current total district count?
- [] **verify**: Any recent development in the Naga political settlement/peace

process (Framework Agreement talks, ceasefire extensions)?

- []  **verify:** Who is the current President, Vice-President, and Chief Justice of India, and has any changed recently?
- []  **verify:** What were the headline announcements of the most recent Union Budget, and what was the latest RBI repo-rate decision?
- []  **verify:** Which major international summit(s) did India host or attend most recently (G20/BRICS/SCO/QUAD/UN), and with what outcome?
- []  **verify:** What was India's most recent major ISRO mission/launch?
- []  **verify:** Who won the most recent Bharat Ratna, Nobel-category, or major national award, if any were announced in the last year?
- []  **verify:** Any recent high-profile diplomatic visit to India or by an Indian leader abroad, and the visiting/visited dignitary's official title (a favourite question type — note how the 2025 paper asked for a visiting official's *exact position*, not just their name).
- []  **verify:** Any major India-related sporting result (medal, series win) in the last 12 months.

Method: build your own one-page current-affairs diary from real sources in the weeks before 26 September, star every Nagaland/NE item, and revise only your own verified notes in the final week — never a guessed fact.

Quick self-test tracker

Section	Questions	Your first-pass score	Re-test score
1. Polity (incl. 371A)	25	___ / 25	___ / 25
2. Nagaland & North-East	35	___ / 35	___ / 35
3. Indian History	20	___ / 20	___ / 20
4. Indian Economy	20	___ / 20	___ / 20
5. Geography	20	___ / 20	___ / 20
6. General Science	20	___ / 20	___ / 20
7. Mental Ability	20	___ / 20	___ / 20
Total scored MCQs	160		

One-line takeaway: this bank gives you 160 original drill questions plus a live current-affairs checklist — your highest-value re-attempts are Section 2 (Nagaland & North-East) and Section 1 (Polity/Article 371A), because that is where a Nagaland candidate's edge is largest and the facts are the most fixed.

● Core — E-R & keys

Source: *_Shared/CORE_06_DBMS.md*

4. The Entity-Relationship (E-R) model

Concept

The **E-R model** (Peter Chen, 1976) is a **high-level conceptual** model used to design a

database *before* choosing tables. It views the world as **entities** (things) linked by **relationships**, each described by **attributes**. The output is an **E-R diagram** — the single most drawable, most examinable artefact in the whole DBMS syllabus. Learn the notation until you can draw it fast.

The building blocks and their symbols

Concept	Symbol (hand-drawn)	Meaning
Entity type	Rectangle	A class of real-world objects: <code>STUDENT</code> , <code>COURSE</code>
Weak entity type	Double rectangle	An entity that cannot be identified by its own attributes alone
Attribute	Ellipse/oval	A property of an entity or relationship: <code>name</code> , <code>age</code>
Key attribute	Ellipse with the name underlined	Uniquely identifies an entity: <code>roll_no</code>
Relationship type	Diamond	An association among entities: <code>ENROLLS</code>
Identifying relationship	Double diamond	Links a weak entity to its owner
Multivalued attribute	Double ellipse	Can hold several values: <code>phone_numbers</code>
Composite attribute	Ellipse with child ellipses	Divisible into parts: <code>address</code> → street, city, pin
Derived attribute	Dashed ellipse	Computed, not stored: <code>age</code> (from <code>dob</code>)
Partial key (of a weak entity)	Dashed underline	Distinguishes weak entities within one owner

Attribute types — a favourite short question

Type	Meaning	Example
Simple (atomic)	Cannot be divided	<code>age</code> , <code>roll_no</code>
Composite	Made of sub-parts	<code>name</code> = first + last; <code>address</code> = street + city + pin
Single-valued	One value per entity	<code>date_of_birth</code>
Multivalued	Several values per entity	<code>phone</code> , <code>email</code>
Stored	Physically stored	<code>date_of_birth</code>
Derived	Computed from others	<code>age</code> from <code>date_of_birth</code>
Key	Uniquely identifies	<code>roll_no</code>
NULL	Value unknown or not applicable	<code>middle_name</code>

Relationships — degree, cardinality, participation

Degree = number of entity types participating in a relationship:

- **Unary (recursive)** — degree 1: `EMPLOYEE manages EMPLOYEE`.
- **Binary** — degree 2 (by far the commonest): `STUDENT enrolls COURSE`.
- **Ternary** — degree 3: `SUPPLIER supplies PART to PROJECT`.

Cardinality ratio = the maximum number of entities one entity can be related to:

Ratio	Meaning	Example
-------	---------	---------

1:1 (one-to-one)	Each A relates to at most one B and vice-versa	EMPLOYEE — manages — DEPARTMENT (one manager per dept)
1:N (one-to-many)	One A relates to many B, each B to one A	DEPARTMENT — has — EMPLOYEE
M:N (many-to-many)	Many A to many B	STUDENT — enrolls — COURSE

Participation constraint = the *minimum* — must every entity take part?

- **Total participation** (existence dependency): drawn as a **double line** — *every* entity of the type must participate. E.g. every LOAN must belong to some CUSTOMER.
- **Partial participation: single line** — participation is optional. E.g. not every CUSTOMER has a LOAN.

Cardinality and participation together are the **(min, max)** notation: (1,1), (0,N), etc. Stating both the ratio *and* the participation is what separates a full answer from a partial one.

Weak entity types

A **weak entity** cannot be uniquely identified by its own attributes; it depends on a **strong (owner) entity** through an **identifying relationship**. It has only a **partial key (discriminator)** — unique only *within* one owner. Classic example: DEPENDENT of an EMPLOYEE. Two dependents of different employees may both be "Sunita, daughter"; a dependent's full key is (employee_id, dependent_name). Draw the weak entity as a **double rectangle**, the identifying relationship as a **double diamond**, with **total participation** (double line) on the weak side, and the partial key **dashed-underlined**.

Enhanced E-R (EER): generalisation, specialisation, aggregation

These three extend the basic model and are explicitly listed in your task — expect a 10-marker.

Specialisation (top-down): start with a general entity (superclass) and split it into more specific subclasses that have extra attributes. EMPLOYEE → ENGINEER, MANAGER, TECHNICIAN. Each subclass **inherits** the superclass's attributes and relationships and adds its own.

Generalisation (bottom-up): the reverse — spot common features among several entities and combine them into a general superclass. CAR and TRUCK → VEHICLE. Generalisation and specialisation are the same "**is-a**" hierarchy approached from opposite directions.

Draw the **is-a** hierarchy as a superclass rectangle connected downward through a **triangle** (labelled "ISA") to its subclass rectangles.

Two constraint dimensions to mention:

- **Disjoint (d)** vs **overlapping (o)**: can an entity belong to *only one* subclass, or several? (An EMPLOYEE is either salaried or hourly = disjoint; a PERSON may be both STUDENT and EMPLOYEE = overlapping.)
- **Total** vs **partial** specialisation: must every superclass member belong to *some* subclass?

Aggregation: a way to express a **relationship of a relationship** — to treat a whole relationship set as a single higher-level entity so that another relationship can connect to it. Classic example: EMPLOYEE works-on PROJECT (a relationship); a MANAGER **supervises** that whole works-on combination. You aggregate works-on into an abstract entity and draw the supervises relationship to the **box drawn around** the aggregated relationship. It solves the problem that basic E-R cannot draw a relationship connected to another relationship.

Abstraction	Direction / idea	Keyword	Example
Specialisation	Top-down: split into subclasses	is-a	EMPLOYEE → MANAGER, ENGINEER
Generalisation	Bottom-up: combine into superclass	is-a	CAR, TRUCK → VEHICLE
Aggregation	Treat a relationship as an entity	part-of / has-a	Manager supervises (Employee works-on Project)

Worked example — a small E-R diagram to describe

Design an E-R schema for a college. Entities and their key attributes:

- **STUDENT** (roll_no, name[composite], dob, age[derived], phone[multivalued])
- **COURSE** (course_id, title, credits)
- **INSTRUCTOR** (emp_id, name, salary)
- **DEPENDENT** (name[partial key], relation) — **weak**, owned by **INSTRUCTOR**

Relationships:

- **ENROLLS** between STUDENT and COURSE — **M:N**, with attribute grade (a relationship attribute — an M:N relationship's own attribute, e.g. the grade a student earns in a course).
- **TEACHES** between INSTRUCTOR and COURSE — **1:N** (one instructor teaches many courses).
- **HAS_DEPENDENT** — identifying relationship (double diamond) between INSTRUCTOR and the weak entity DEPENDENT, total participation on DEPENDENT.

Describe it in words for the examiner, then draw: rectangles for the four entities, a double rectangle for DEPENDENT, diamonds for the relationships, ellipses for attributes with roll_no, course_id, emp_id underlined, age dashed, phone a double ellipse, name composite, and grade hanging off the ENROLLS diamond.

Likely exam questions

- **[5]** Define entity, attribute and relationship. List the different types of attribute.
- **[5]** Explain the different types of attribute with examples and their E-R symbols.
- **[5]** What is a weak entity? How is it represented? Give an example.
- **[5]** Differentiate between generalisation, specialisation and aggregation.
- **[5]** Explain cardinality ratios and participation constraints with examples.
- **[10]** Explain the E-R model. Describe all the symbols used in an E-R diagram with examples.
- **[15]** Draw an E-R diagram for a [college / hospital / bank] system. Show entities, attributes (including composite, multivalued, derived and key), relationships with their cardinality and participation, at least one weak entity, and one generalisation hierarchy.

MCQ traps

Trap	Correct answer
"An entity set is drawn as a?"	Rectangle . Relationship = diamond , attribute = ellipse .
"A derived attribute is drawn as?"	Dashed ellipse . Multivalued = double ellipse .
"A weak entity is drawn as?"	Double rectangle ; identifying relationship = double diamond .
"Total participation is shown by?"	A double line .
"Degree of a relationship means?"	Number of entity types in it.
"Combining lower entities into a higher one is?"	Generalisation (bottom-up).

"Relationship treated as an entity is?"	Aggregation .
"'is-a' relationship denotes?"	Generalisation/specialisation (inheritance).
"Who proposed the E-R model?"	Peter Chen (1976) .

5. Converting an E-R diagram to a relational schema

Concept

Design in E-R (easy to think in), then **mechanically translate** to tables (easy to implement).
 There are **seven standard rules**; knowing all seven and applying them to a given diagram is a classic 10/15-mark question named in *all three* of your syllabi.

#	E-R construct	Relational rule
1	Strong (regular) entity	Create a table with all its simple attributes; its key becomes the primary key
2	Weak entity	Create a table with its own attributes plus the primary key of the owner as a foreign key; the primary key = owner's PK + the partial key
3	1:1 relationship	Add the PK of one side into the other as a foreign key (put it on the side with total participation to avoid NULLs). No new table needed
4	1:N relationship	Add the PK of the " 1 " side into the " N " side table as a foreign key. No new table
5	M:N relationship	Create a new (junction/bridge) table containing the PKs of both entities as foreign keys; their combination is the primary key , plus any relationship attributes
6	Multivalued attribute	Create a separate table holding the attribute + the owner's PK; PK = both together
7	Composite attribute	Store only its simple components as separate columns; drop the composite name itself

For **generalisation/specialisation**, three strategies: (a) one table per subclass with the inherited attributes duplicated; (b) one table for the superclass and one per subclass sharing the key; (c) a single table for the whole hierarchy with a type-discriminator column and NULLs for inapplicable attributes.

Worked example

Take the college E-R model from §4 and translate it.

Rule 1 — strong entities:

```
STUDENT(roll_no PK, first_name, last_name, dob)      -- age is derived → not stored
COURSE(course_id PK, title, credits)
INSTRUCTOR(emp_id PK, name, salary)
```

name was composite → split into **first_name**, **last_name** (Rule 7). **age** was derived → not stored (compute on demand).

Rule 6 — multivalued attribute **phone** of STUDENT:

```
STUDENT_PHONE(roll_no FK, phone)    PK = (roll_no, phone)
```

Rule 4 — 1:N relationship TEACHES (one instructor, many courses): put the instructor's key into COURSE:

```
COURSE(course_id PK, title, credits, emp_id FK → INSTRUCTOR)
```

Rule 5 — M:N relationship ENROLLS with attribute **grade**: make a junction table:

```
ENROLLS(roll_no FK → STUDENT, course_id FK → COURSE, grade)    PK = (roll_no, course_id)
```

Rule 2 — weak entity DEPENDENT owned by INSTRUCTOR:

```
DEPENDENT(emp_id FK → INSTRUCTOR, dep_name, relation)    PK = (emp_id, dep_name)
```

Final relational schema:

```
STUDENT(roll_no, first_name, last_name, dob)
STUDENT_PHONE(roll_no, phone)
COURSE(course_id, title, credits, emp_id)
INSTRUCTOR(emp_id, name, salary)
ENROLLS(roll_no, course_id, grade)
DEPENDENT(emp_id, dep_name, relation)
```

Note how the M:N **ENROLLS** became its own table (Rule 5) while the 1:N **TEACHES** needed only a foreign column (Rule 4) — **that difference is the most examined point in the whole topic.**

Likely exam questions

- **[5]** State the rules for converting an E-R diagram into relational tables.
- **[5]** How are 1:N and M:N relationships mapped to tables? Why do they differ?
- **[10]** Given an E-R diagram, convert it into a set of relational schemas, showing primary and foreign keys, and justify each step.

MCQ traps

Trap	Correct answer
"An M:N relationship becomes?"	A separate (junction) table with both PKs.
"A 1:N relationship is mapped by?"	Putting the " 1 "-side PK as an FK in the "N"-side table.
"A multivalued attribute becomes?"	A separate table .
"A weak entity's PK is?"	Owner's PK + its own partial key .
"A composite attribute is stored as?"	Its individual simple components .

6. The relational model — terminology

Concept

A **relation** is a table; the model rests on a handful of precise terms that examiners love to test because each maps to a plain word.

Formal term	Informal term	Meaning
Relation	Table	A set of tuples over the same attributes
Tuple	Row / record	One entity's data
Attribute	Column / field	A named property

Domain	—	The set of allowed atomic values for an attribute (e.g. <code>age</code> : integers 0–150)
Degree (arity)	—	The number of attributes (columns)
Cardinality	—	The number of tuples (rows)
Relation schema	Table structure	<code>STUDENT(roll_no, name, age)</code> — name + attributes
Relation instance	Table contents	The actual set of rows at a moment

Do not confuse degree and cardinality: degree counts **columns**, cardinality counts **rows**. A table `STUDENT(roll_no, name, age, dept)` with 50 students has **degree 4** and **cardinality 50**.

Properties of a relation (Codd)

- Each **tuple is unique** — no duplicate rows (a relation is a *set*).
- **Tuple order is immaterial** — rows have no inherent sequence.
- **Attribute order is immaterial** — columns are identified by name, not position.
- Each **cell holds a single atomic value** — this is the **first normal form** requirement; no repeating groups or multivalued cells.
- All values in a column come from the **same domain**.
- **NULL** marks an unknown or inapplicable value.

Likely exam questions

- [5] Define relation, tuple, attribute, domain, degree and cardinality with an example.
- [5] State the properties of a relation.
- [5] Differentiate between degree and cardinality of a relation.

MCQ traps

Trap	Correct answer
"Number of columns is the?"	Degree . Number of rows = cardinality .
"A row is formally a?"	Tuple .
"The set of permitted values for an attribute?"	Domain .
"A relation is a set of?"	Tuples — hence no duplicate rows.
"Order of rows in a relation matters?"	No .
"Every cell must be?"	Atomic (single-valued) — 1NF.

7. Keys

Concept

A **key** is one or more attributes that let you **identify tuples**. Everything in normalization and integrity builds on keys, so get the hierarchy exact. Work from a single running example throughout:

```
STUDENT(roll_no, aadhaar, email, name, dept)
-- roll_no unique, aadhaar unique, email unique
```

Key	Definition	In the example
Super key	Any set of attributes that uniquely identifies a tuple (may have extras)	{roll_no}, {roll_no, name}, {aadhaar}, {roll_no, aadhaar} ... many
Candidate key	A minimal super key — no attribute can be removed and still stay unique	{roll_no}, {aadhaar}, {email}
Primary key	The one candidate key chosen to identify tuples; cannot be NULL, must be unique	roll_no (chosen)
Alternate key	Candidate keys not chosen as primary	aadhaar, email
Composite (concatenated) key	A key made of two or more attributes together	In ENROLLS(roll_no, course_id, grade): (roll_no, course_id)
Foreign key	An attribute in one relation that refers to the primary key of another (may repeat, may be NULL)	emp_id in COURSE referencing INSTRUCTOR
Surrogate key	A system-generated artificial key (auto-increment/UUID) with no business meaning	An added student_id INT AUTO_INCREMENT

The containment chain to state: every **primary key** and every **alternate key** is a **candidate key**; every candidate key is a **super key**. So $\text{super key} \supseteq \text{candidate key} \supseteq \{\text{primary key, alternate keys}\}$. A super key with any extra attribute removed that is still unique was not minimal — minimality is exactly what turns a super key into a candidate key.

Foreign key rules (referential integrity): a foreign-key value must either **match an existing primary-key value** in the referenced table **or be NULL**. On delete/update of the parent, the DBMS applies a chosen action: **CASCADE, SET NULL, SET DEFAULT, RESTRICT/NO ACTION**.

Likely exam questions

- **[5]** Define super key, candidate key, primary key and alternate key with one example.
- **[5]** What is a foreign key? How does it enforce referential integrity?
- **[5]** Differentiate between a primary key and a candidate key.
- **[10]** Explain the various types of keys in the relational model with a single worked example, and show the relationship among them.

MCQ traps

Trap	Correct answer
"A minimal super key is a?"	Candidate key.
"Candidate keys not chosen as primary are?"	Alternate keys.
"Which key cannot be NULL?"	Primary key (entity integrity).
"Which key can repeat and be NULL?"	Foreign key.
"A key of two or more attributes is?"	Composite key.
"Every candidate key is a?"	Super key (the reverse is not true).
"Auto-generated meaningless key?"	Surrogate key.

17. Distributed databases, the network model and other models

Concept — Distributed databases (DDB)

A **distributed database** is a single logical database whose data is **stored across multiple physical sites** connected by a network, managed by a **DDBMS** that makes the distribution **transparent** to the user (it looks like one database).

Key techniques:

- **Fragmentation** — splitting a table: **horizontal** (rows, e.g. by region), **vertical** (columns), or **hybrid**.
- **Replication** — keeping copies of data at several sites for **availability and read performance**, at the cost of **consistency maintenance** on writes.
- **Allocation** — deciding which fragment/replica lives where.

Transparencies to name: location, fragmentation, replication, and concurrency/failure transparency.

Advantages	Disadvantages
Local autonomy and faster local access	Complexity of design and management
Reliability/availability — one site down, others serve	Costly concurrency control and recovery across sites
Scalability — add sites incrementally	Security harder over a network
Matches geographically distributed organisations	Distributed commit needed

Distributed commit uses the **two-phase commit (2PC)** protocol: a coordinator asks all sites to **prepare** (phase 1); only if **all** vote "yes" does it broadcast **commit** (phase 2), otherwise **abort** — guaranteeing atomicity across sites. (Contrast the earlier **2PL**, which is about locking, not commit — a very common confusion.)

CAP theorem (worth a line): a distributed system can guarantee at most two of **Consistency**, **Availability**, **Partition-tolerance** simultaneously.

The network model (named in the CS Degree syllabus)

The **network data model** (CODASYL/DBTG, 1971) represents data as a **graph of records connected by links called sets** (an owner–member relationship). Unlike the hierarchical model's tree, a **member record can have several owners**, so it can represent **M:N** relationships more naturally. Navigation follows **pointer chains**, so access is **procedural**.

	Hierarchical	Network	Relational
Structure	Tree	Graph	Tables
Parents per record	One	Many	(values, not links)
Relationships	1:N	1:N and M:N	via foreign keys
Navigation	Procedural (pointers)	Procedural (pointers)	Declarative (SQL)
Example	IBM IMS	CODASYL, IDMS	Oracle, MySQL

Advantages of the network model: represents M:N naturally, efficient navigation, enforces integrity via set membership. **Disadvantages:** structurally complex, no ad-hoc querying, programs are tied to the physical pointer structure (poor data independence) — which is exactly why the relational model displaced it.

Object and object-relational databases (named in the Forensic syllabus)

- **OODBMS** stores **objects** directly (attributes + methods + inheritance + object identity), good for complex data (CAD, GIS, multimedia) and eliminating the object–relational "impedance mismatch," but lacks a standard query language as mature as SQL.
- **ORDBMS** extends the relational model with **user-defined types, inheritance, and complex attributes** while keeping SQL — the pragmatic middle ground (PostgreSQL, Oracle).

Likely exam questions

- **[5]** What is a distributed database? State its advantages and disadvantages.
- **[5]** Differentiate horizontal and vertical fragmentation. What is replication?
- **[5]** Explain the network data model with an example.
- **[5]** Differentiate the hierarchical, network and relational models.
- **[10]** Explain distributed databases: fragmentation, replication, transparency and the two-phase commit protocol.

MCQ traps

Trap	Correct answer
"Splitting a table by rows is?"	Horizontal fragmentation . By columns = vertical .
"Keeping copies at several sites is?"	Replication .
"Distributed atomic commit uses?"	Two-phase commit (2PC) — not 2PL.
"The network model is also called?"	CODASYL / DBTG .
"Which model allows a record many owners?"	Network .
"CAP theorem — you can guarantee at most?"	Two of C, A, P .
"ORDBMS keeps which query language?"	SQL (extended).